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SECR1013 DIGITAL LOGIC

MODULE 2a: NUMBER SYSTEMS

2022/2023-1

FACULTY OF COMPUTING

Extra

Exercise 2a.1:

Express 567.23_{10} as a sum of values of each digit.

Exercise 2a.2:

Express 110100.011_2 as a sum of values of each digit.

3

Exercise 2a.3:

Express 568.23_8 as a sum of values of each digit.

4

Exercise 2a.3:

Express 567.23_8 as a sum of values of each digit.

5

Exercise 2a.4:

Express 567.23_{16} as a sum of values of each digit.

6

Exercise 2a.4b:

Express $5A7.2F_{16}$ as a sum of values of each digit.

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Calculate the value in decimal to all previous exercise.

Exercise 2a.2:
110100.011_2

Exercise 2a.3:
567.23_8

Exercise 2a.4:
567.23_{16}

Exercise 2a.4b:
$5A7.2F_{16}$

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Exercise 2a.4c:

Calculate the value of 42.03_5 as decimal number.

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Exercise 2a.5:

Simple Deduction: Binary Number

- Fill in the blank spaces.

(a)			(b)		(c)	
Binary	**Decimal**	2^x	**Binary**	**Decimal**	**Binary**	**Decimal**
	2			1	101	5
	4			3		10
1000	8	2^3		7		17
	16			15		26
	32		11111	31		35

Solution:

ii. Based on the answers that you have calculated in the table, select the correct answer below that best described the deduced observation for (a), (b) and (c).

- ☐ An even number will have a zero as the last bit while an odd number will have a one as the last bit.
- ☐ The power of two is equivalent to the number of zeroes in the binary representation number.
- ☐ A binary number that is equal to $2^x - 1$, will consist of all ones.

Extra

Exercise 2a.6:

1447_{10} = _____ 16

(inverted from exercise 4b)

13

Extra

Exercise 2a.7:

1447.18359_{10} = _____ 16

Successive Division: (Whole part)	Successive Multiplication: (Fraction part)
	(up to 3 fractional points)

(inverted from exercise 4b)

14

Extra

Exercise 2a.8:

$AF90.7B_{16}$ = _____ 2

15

Extra

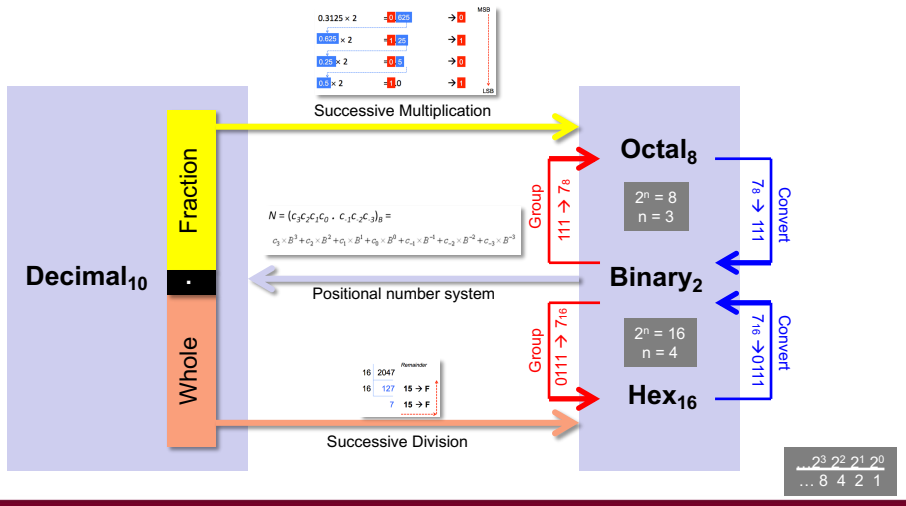
Exercise 2a.9:

45.27_{8} = _____ 2

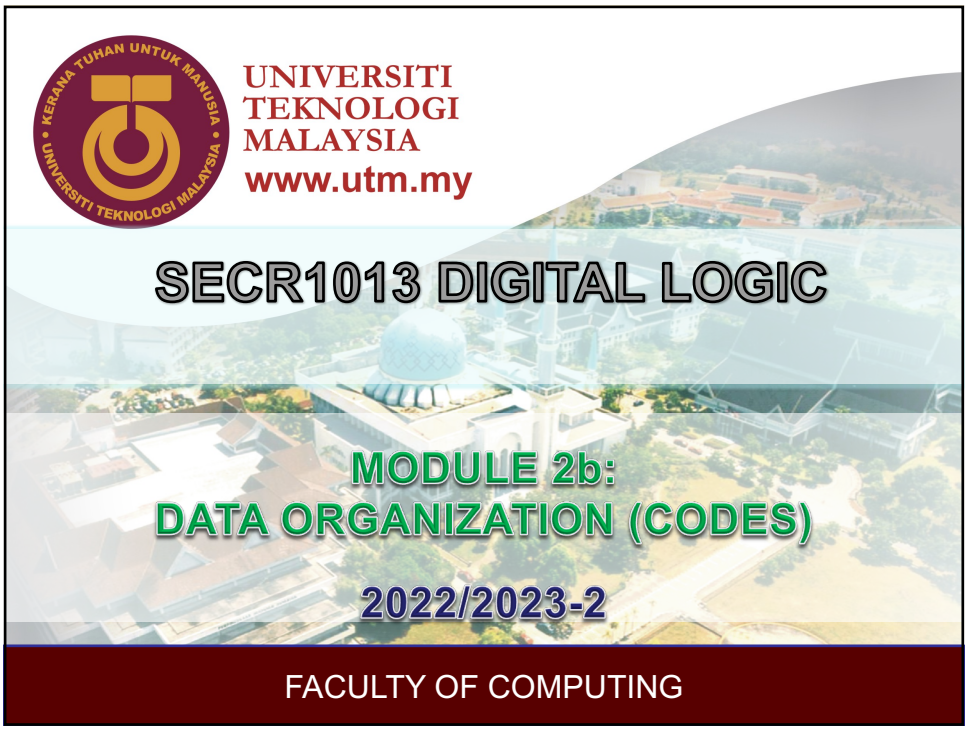
16

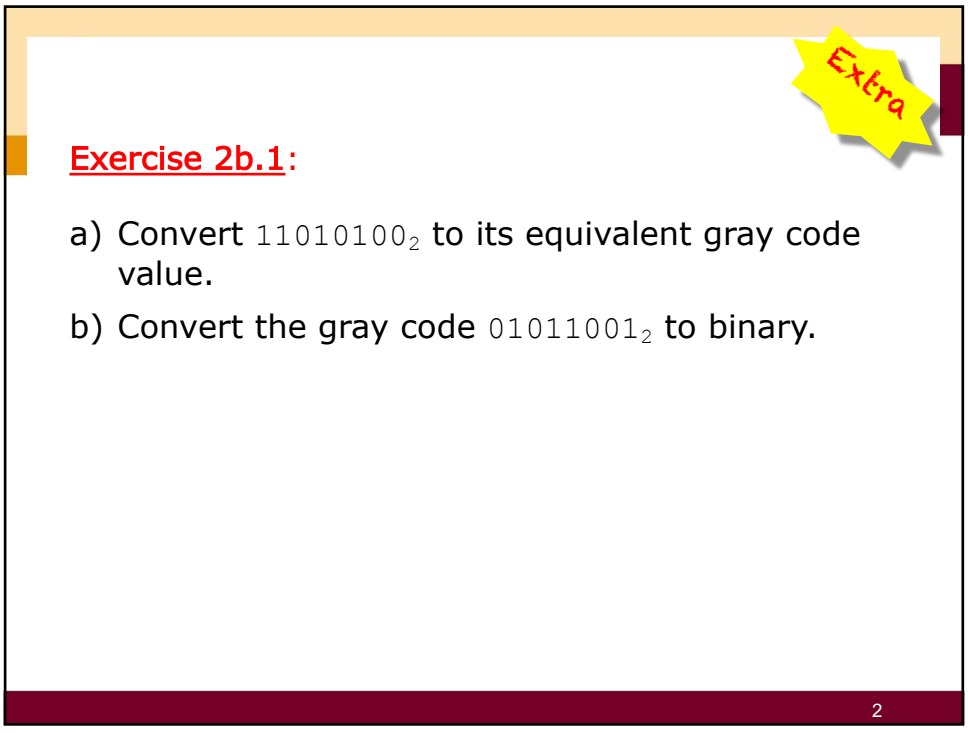
Extra

Summary of Number Systems Conversion



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MODULE 2b:
DATA ORGANIZATION (CODES)
2022/2023-2
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Extra
Exercise 2b.1:
a) Convert 11010100_2 to its equivalent gray code value.
b) Convert the gray code 01011001_2 to binary.
2

Decimal	Hex	ASCII	Decimal	Hex	ASCII	Decimal	Hex	ASCII	Decimal	Hex	ASCII
0	00	NUL	32	20	(blank)	64	40	@	96	60	`
1	01	SOH	33	21	!	65	41	A	97	61	a
2	02	STX	34	22	"	66	42	B	98	62	b
3	03	ETX	35	23	#	67	43	C	99	63	c
4	04	EOT	36	24	$	68	44	D	100	64	d
5	05	ENQ	37	25	%	69	45	E	101	65	e
6	06	ACK	38	26	&	70	46	F	102	66	f
7	07	BEL	39	27	'	71	47	G	103	67	g
8	08	BS	40	28	(	72	48	H	104	68	h
9	09	HT	41	29	)	73	49	I	105	69	i
10	0A	LF	42	2A	*	74	4A	J	106	6A	j
11	0B	VT	43	2B	+	75	4B	K	107	6B	k
12	0C	FF	44	2C	,	76	4C	L	108	6C	l
13	0D	CR	45	2D	-	77	4D	M	109	6D	m
14	0E	SO	46	2E	.	78	4E	N	110	6E	n
15	0F	SI	47	2F	/	79	4F	O	111	6F	o
16	10	DLE	48	30	0	80	50	P	112	70	p
17	11	DC1	49	31	1	81	51	Q	113	71	q
18	12	DC2	50	32	2	82	52	R	114	72	r
19	13	DC3	51	33	3	83	53	S	115	73	s
20	14	DC4	52	34	4	84	54	T	116	74	t
21	15	NAK	53	35	5	85	55	U	117	75	u
22	16	SYN	54	36	6	86	56	V	118	76	v
23	17	ETB	55	37	7	87	57	W	119	77	w
24	18	CAN	56	38	8	88	58	X	120	78	x
25	19	EM	57	39	9	89	59	Y	121	79	y
26	1A	SUB	58	3A	:	90	5A	Z	122	7A	z
27	1B	ESC	59	3B	;	91	5B	[	123	7B	{
28	1C	FS	60	3C	<	92	5C	\	124	7C	|
29	1D	GS	61	3D	=	93	5D	]	125	7D	}
30	1E	RS	62	3E	>	94	5E	^	126	7E	~
31	1F	US	63	3F	?	95	5F	_	127	7F	(delete)

Resource: http://www.s4a.us/support/cimco_faq.htm

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Exercise 2b.2:

Convert the string SCR1013 to its ASCII hexadecimal value.

SCR1013 = 53 43 52 31 30 31 33

By using even parity coding, calculate the parity bit and insert this bit at the MSB position. Recalculate the ASCII value in its hexadecimal representation.

Continue...

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Extra

Exercise 2b.3:

Given a string UTM1435h.

a) Convert the string to its ASCII hexadecimal value.

b) Calculate the odd parity bit and insert as MSB.

a) Recalculate the ASCII value in hexadecimal.

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Number 1s	Even Parity	Odd Parity
Even	0	1
Odd	1	0

Extra

Character (ASCII)	ASCII (Hex)	Binary	Odd parity bit + Binary	New ASCII (Hex)
U				
T				
M				
1				
4				
3				
5				
h				

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MODULE 2c: ARITHMETIC OPERATIONS

2022/2023-2

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Extra

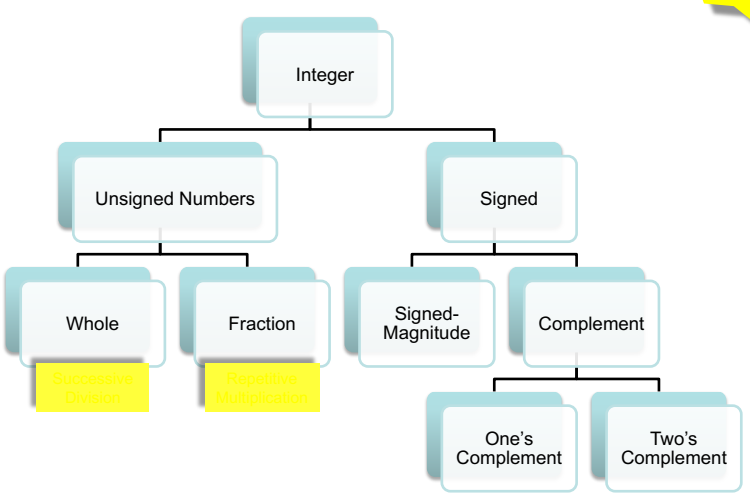
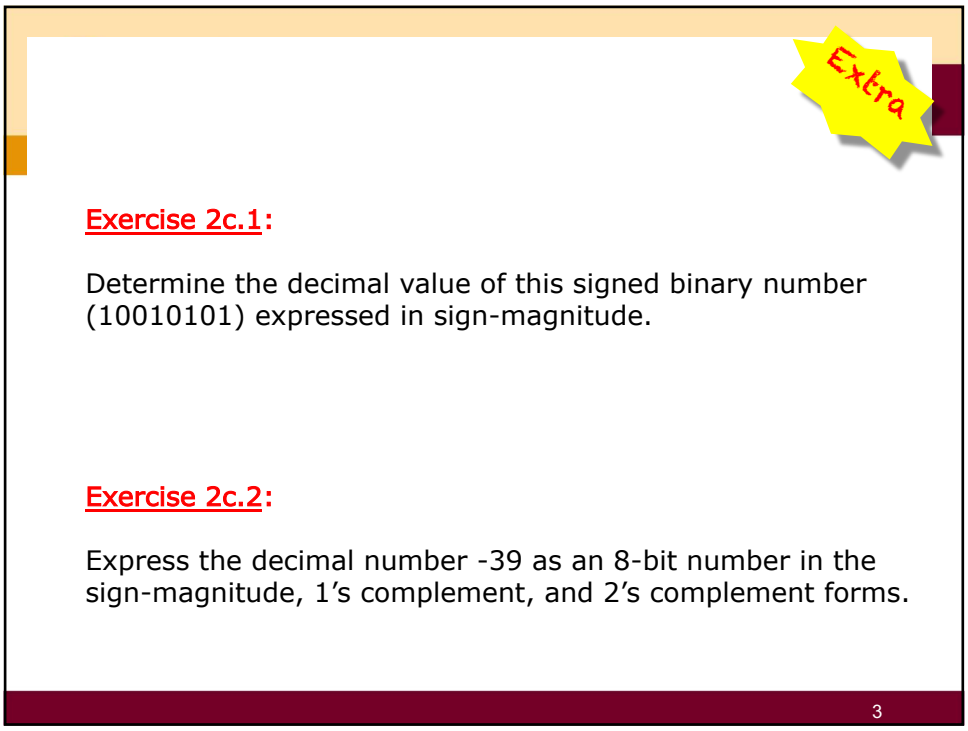


Figure: Summary of Integer Representation

2


Extra
Exercise 2c.1:
Determine the decimal value of this signed binary number (10010101) expressed in sign-magnitude.
Exercise 2c.2:
Express the decimal number -39 as an 8-bit number in the sign-magnitude, 1's complement, and 2's complement forms.
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